Data Cleaning & Validation Report

Week 2 — Opportunity Data & Learner Data (Excelerate)

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1. Data Integration Process

Two levels of integration were investigated: a record-level join (linking an individual learner record to a specific opportunity record) and, where that was not possible, a category-level integration (combining both tables on a shared dimension).

1.1 Record-Level Join — Investigated and Ruled Out

Seven candidate fields were systematically tested as a possible record-level link between Learner Data and Opportunity Data:

Candidate Field	Tested Against	Result
relation_id	opportunity_id, pk, and all embedded JSON reference fields	0 matches. Only 38 of 1,979 records even have a non-null relation_id.
work_item_sk	opportunity_data.pk	28,660 "matches" — false positive caused by pk not being unique (mostly the literal text "Opportunity#").
work_item_sk	opportunity_data.opportunity_id	15 matches — false positive, traced to the small number of corrupted rows with shifted column data.
cohort_p1 / cohort_p2 / assigned_cohort	All embedded JSON reference fields in Opportunity Data	0 matches against any field.
external_reference_url	Manual inspection	Contains page-navigation URLs, not opportunity references.
code	learner_data.application_id	0 matches.
code	learner_data.transaction_id	0 matches.

Conclusion: no reliable field exists in either export that links a specific Learner Data record to a specific Opportunity Data record. Every apparent match was either a false positive caused by data corruption/non-unique keys, or returned zero results. A true record-level join would require a dedicated bridge export from the source system (true opportunity_id-to-application_id pairs) that was not available for this task.

1.2 Category-Level Integration — Implemented

Both tables share a category field, which was investigated as a basis for logical (rather than record-level) integration. An initial join attempt returned zero matching records despite both tables containing visibly identical category values (Internship, Career, Competition, etc.).

Investigation traced this to hidden leading/trailing whitespace in the category field — the values matched to the eye but not under a strict equality comparison.

The join logic was corrected by trimming whitespace from both sides before comparing, and by aggregating each table independently by category before joining the two summaries. This avoids a row-multiplication problem that occurs when joining directly on a non-unique field (an initial attempt produced 612,917 rows, since category is shared by many records on both sides).

Join Logic Used

```
CREATE VIEW integrated_category_summary AS
SELECT o.cat AS category, o.total_opportunities, o.avg_opportunity_fee,
      l.total_learner_records, l.learner_applications, l.learner_completions
FROM (
  SELECT TRIM(category) AS cat,
        COUNT(DISTINCT opportunity_id) AS total_opportunities,
        ROUND(AVG(fee), 2) AS avg_opportunity_fee
  FROM opportunity_data_clean GROUP BY TRIM(category)
) o
LEFT JOIN (
  SELECT TRIM(category) AS cat,
        COUNT(*) AS total_learner_records,
        COUNT(DISTINCT application_id) AS learner_applications,
        COUNT(completion_date) AS learner_completions
  FROM learner_data_clean GROUP BY TRIM(category)
) l ON o.cat = l.cat
ORDER BY o.total_opportunities DESC;
```

Key Fields Involved

- Join key: category (trimmed, matched exactly across both tables)
- Opportunity Data aggregates: opportunity_id (count, distinct), fee (average)
- Learner Data aggregates: total record count, application_id (distinct, non-null), completion_date (non-null count)

Structure of the Integrated Dataset

The resulting integrated_category_summary view has one row per category (11 rows) and combines metrics from both source tables side by side:

category	total_opportunities	avg_opportunity_fee	total_learner_records	learner_applications	learner_completions
Internship	1591	60.97	184	0	0
Career	1087	184.86	129	0	0
Competition	734	1.72	66	0	0
Course	631	39.52	68	0	0
Event	621	4.86	78	0	0
Engagement	459	1.36	44	0	0
Masterclass	400	1.26	41	0	0

JobSimulation	198	2.89	18	0	0

Note: learner_applications and learner_completions show 0 in every category. Validation queries showed why: category is missing for 1,351 of the 1,979 Learner Data records (68%), and all 203 records with a usable application_id and all 162 records with a completion_date fall within that uncategorised group. The category-level integration therefore covers 628 learner records (32%). The zeros reflect the source data, not a fault in the join.

2. Data Cleaning Activities

Cleaning was performed on working copies of each table (opportunity_data_clean and learner_data_clean), leaving the original imported tables untouched for comparison.

2.1 Opportunity Data — Issues Identified and Actions Taken

Issue	Action Taken	Rationale
Corrupted rows (JSON fragments in fee, currency_type, is_auto_approve, is_archived, duration_type)	Rows removed (6 records)	Original correct values cannot be recovered from the data itself.
Literal text "NULL" instead of true empty values	Converted to true SQL NULL	A cell containing the word "NULL" is a missing value, not a category, and was breaking type conversion and skewing counts.
Inconsistent location values ("Virtual"/"virtual"/"vitrual", "Work From Home"/"WFM"/"wfm")	Standardised to single consistent labels	Same real-world value was being split across multiple categories due to case/spelling differences.
Duplicate currency category ("EUR" vs "EURO")	Merged into "EUR"	Same currency represented two ways.
Singular/plural and typo inconsistency in duration_type	Standardised to plural form	Consolidates genuinely identical duration units.
fee stored as text	Converted to numeric	Required for aggregation and statistical analysis.
Date fields stored as millisecond epoch timestamps (text)	Converted to proper timestamp type	Required for date-based filtering, sorting, and trend analysis.
is_archived, is_auto_approve stored as text	Converted to boolean	Enables correct TRUE/FALSE filtering.

Hidden whitespace in category field (discovered during integration work)	Trimmed using TRIM() at query time	Caused a complete join failure despite visually identical values; trimming is applied wherever category is used for grouping or joining.

Fields left unchanged: Reward, CareerAddOn, Cohort, Eligibility, NotStartedTransaction, DropoutTransaction, Panellist, Testimonial, and tracking_questions remain embedded JSON and were judged out of scope for this cleaning pass.

2.2 Learner Data — Issues Identified and Actions Taken

Issue	Action Taken	Rationale
Inconsistent lead source values	Standardised to single consistent labels	Same lead source was being split into duplicate categories.
application_id containing obvious test/placeholder values	Set to NULL	Not genuine identifiers; would corrupt counts or joins. Only 85 distinct values existed across the column to begin with.
Literal text "NULL" across date, fee, and amount fields	Converted to true SQL NULL	Consistent with the same fix applied to Opportunity Data.
Mixed date formats (ISO text and millisecond epoch)	Converted to proper timestamp type	Required for trend analysis across sign-up, completion, and withdrawal dates.
fee and amount_to_be_paid stored as text, with isolated corrupted values (1 in dropped_out_date, 43 in accept_reject_date)	Corrupted values set to NULL, remainder converted to numeric/timestamp	Isolated corruption, inconsistent with surrounding clean data.
amount_to_be_paid contained impossible values (8 rows above 10,000, including five at 6,000,000,000,000)	Set to NULL	The values were far above the next largest real amounts (2,000 and below) and look like test or mistyped entries. The 10,000 cut-off is a judgement call, recorded here as an assumption. The column total fell from about 30 trillion to 14,234.
Literal text “null” in the category field	Converted to true SQL NULL	Consistent with the treatment of other columns, and allows missing categories to be counted correctly (1,351 records).
status field uses undocumented numeric codes	Left as-is; documented as a known limitation	No lookup/reference table was supplied with this dataset.

212 of 256 columns unnamed (col_45–col_256)	Left as-is; documented as a known limitation	Content and meaning unknown; renaming without source documentation would risk mislabeling data.

3. Validation & Comparison

Metric	Before Cleaning	After Cleaning	Change
Opportunity Data — total records	5,733	5,727	6 corrupted rows removed (0.1%)
Opportunity Data — distinct location categories	23	14	9 duplicate categories consolidated
Opportunity Data — distinct duration_type categories	18	6	12 duplicate/typo categories consolidated
Learner Data — total records	1,979	1,979	No records removed — only values cleaned
Learner Data — distinct lead source categories	14	10	4 duplicate categories consolidated
Learner Data — amount_to_be_paid total	About 30 trillion	14,234	8 invalid values removed
Category-level join (Opportunity to Learner)	0 matches (untrimmed)	11 categories successfully matched (trimmed)	Whitespace fix enabled integration

Validation Checks Performed

- Row counts re-verified after every cleaning stage to confirm no unintended data loss.
- Data type conversions verified via information_schema.columns after each ALTER TABLE statement.
- Category join tested at three levels of strictness (exact match, TRIM, TRIM+LOWER) to isolate the root cause of the initial 0-match result.
- Re-ran category distribution queries after each standardisation step to confirm duplicate categories had merged correctly.

Key Improvement

Beyond the categorical consistency gains (location: 23→14, duration_type: 18→6, lead source: 14→10), the most significant Week 2 improvement was diagnosing and fixing the hidden

whitespace issue in category — without this, a seemingly reasonable integration query would have silently returned zero useful rows, with no error to signal the problem.

4. Updated Analysis

Sample queries below demonstrate the cleaned, integrated dataset is ready for dashboard development.

4.1 Category-Level Integrated Summary

```
SELECT * FROM integrated_category_summary;
```

4.2 Fee Summary Statistics by Currency

```
SELECT currency_type, COUNT(*) AS total_opportunities,
  ROUND(AVG(fee), 2) AS avg_fee, MIN(fee) AS min_fee, MAX(fee) AS max_fee
FROM opportunity_data_clean
WHERE fee IS NOT NULL
GROUP BY currency_type;
```

4.3 Application Trend Over Time

```
SELECT DATE_TRUNC('month', apply_date) AS application_month, COUNT(*) AS total_applications
FROM learner_data_clean
WHERE apply_date IS NOT NULL
GROUP BY application_month
ORDER BY application_month;
```

4.4 Completion Funnel Summary

```
SELECT COUNT(*) AS total_records, COUNT(completion_date) AS completed,
  COUNT(dropped_out_date) AS dropped_out, COUNT(withdraw_date) AS withdrawn,
  COUNT(not_started_date) AS not_started
FROM learner_data_clean;
```

4.5 Standardised Lead Source Performance

```
SELECT from_where_did_you_hear_about_us, COUNT(*) AS total
FROM learner_data_clean
GROUP BY from_where_did_you_hear_about_us
ORDER BY total DESC;
```

These queries directly support the Week 3 dashboard requirements: opportunity sign-up performance, reward/fee summary, application trends over time, completion rates, and lead source performance.

Known Limitations Carried Forward

- Integration is category-level only; no record-level link between a specific learner and a specific opportunity is possible with the current exports.

- Category is missing for 1,351 of 1,979 learner records (68%), so the category-level integration covers only 628 learner records. All completion dates and all usable application IDs fall in the uncategorised group.
- The Reward field and eight other Opportunity Data columns remain embedded JSON and are not usable for numeric analysis without further parsing.
- 212 of 256 Learner Data columns remain unnamed and of unknown content.
- The status field's numeric codes remain undecoded without a reference/lookup table.
- application_id remains unreliable even after cleaning: only 203 of 1,979 records carry a usable value.